

Customer Churn Prediction

Exploratory Data Analysis Report

EDA + Visualizations + Key Insights

Role: Data Analyst | Dataset: 999,999 Rows | 32 Columns

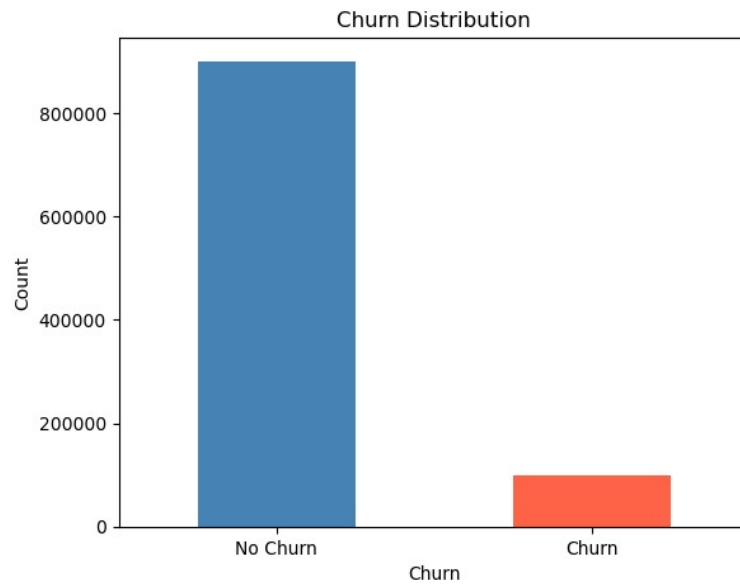
1. Dataset Overview

The Customer Churn dataset contains one million records of telecom customers. The goal of this analysis is to uncover patterns that explain why customers leave.

Metric	Value
Total Customers	999,999
Churned Customers	99,227 (9.9%)
Retained Customers	900,772 (90.1%)
Total Features	32 columns
Missing Values	5 columns — all under 5%

2. Churn Distribution

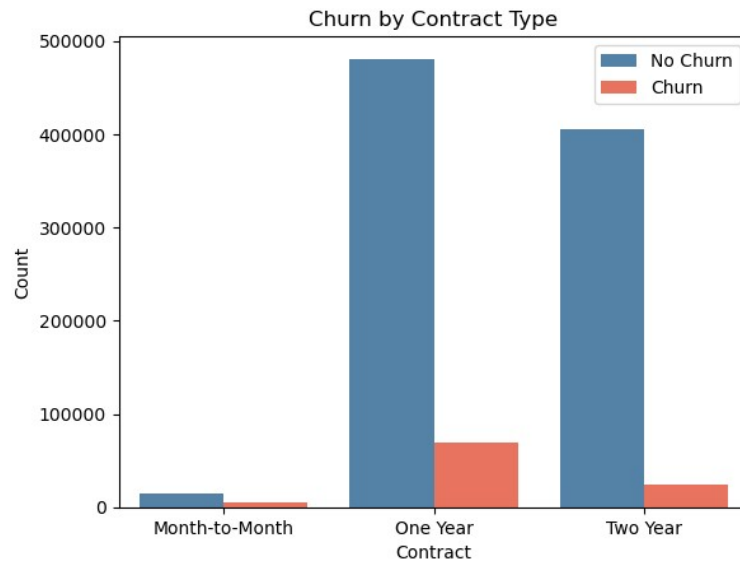
The dataset shows a significant class imbalance. Only 9.9% of customers churned, while 90.1% remained.



💡 Class Imbalance of 9:1 means accuracy alone is misleading. F1-Score and Recall are more meaningful metrics.

3. Churn by Contract Type

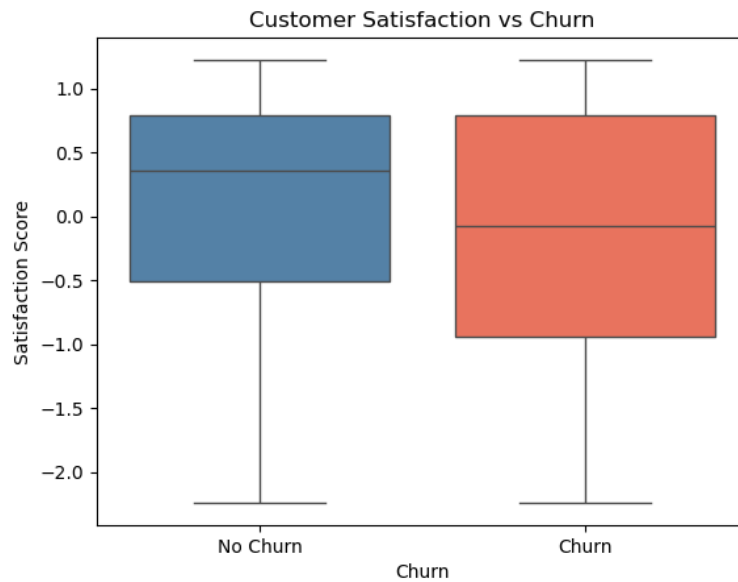
Contract type is one of the strongest predictors of churn. Customers on flexible monthly contracts are more likely to leave.



💡 Month-to-Month customers have the highest churn. Two-Year contract customers are the most loyal.

4. Customer Satisfaction vs Churn

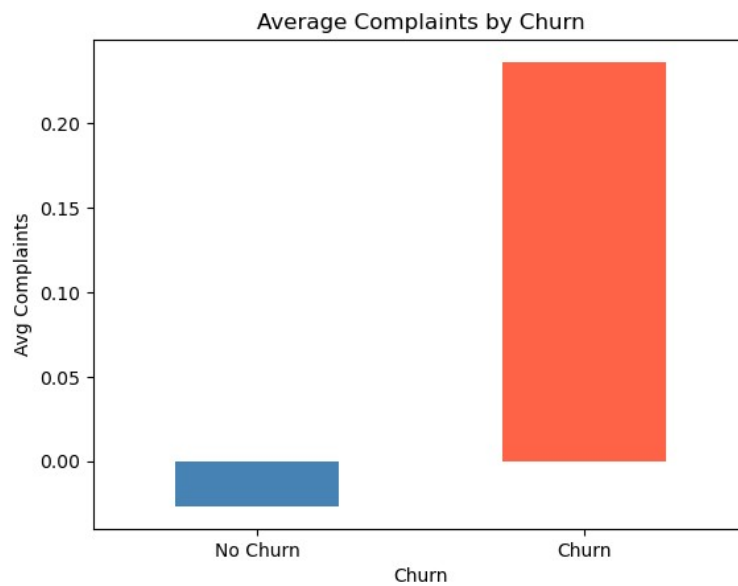
Customer satisfaction scores show a clear negative correlation with churn. Customers who left had lower satisfaction scores.



💡 Churned customers have lower median satisfaction. Satisfaction score is a strong early warning signal.

5. Average Complaints vs Churn

Customers who churned submitted significantly more complaints on average than those who stayed.



💡 Churned customers average more complaints than retained ones. Monitoring complaints is a key retention signal.

6. Correlation Heatmap

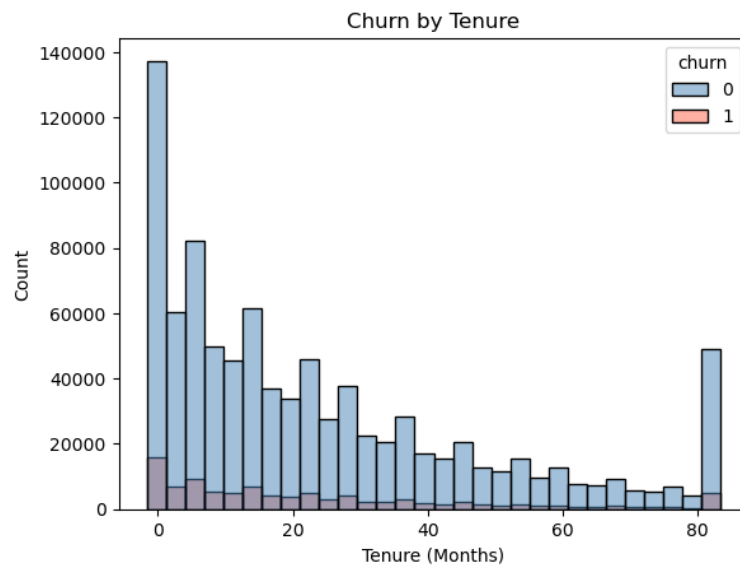
The heatmap shows pairwise correlations between all numerical features and the churn target.



Feature	Correlation	Meaning
num_complaints	+0.08	More complaints → more likely to churn
num_service_calls	+0.08	More service calls → more likely to churn
customer_satisfaction	-0.08	Lower satisfaction → more likely to churn
late_payments	+0.05	More late payments → slight churn risk

7. Churn by Tenure

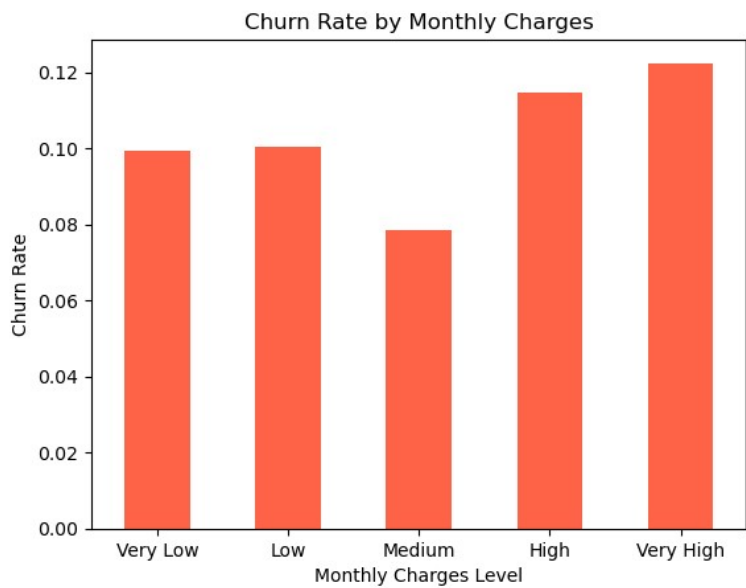
New customers in their first 10 months are far more likely to churn. Long-term customers (40+ months) show strong loyalty.



💡 Highest churn occurs in the first 10 months. Early customer engagement programs could significantly reduce churn.

8. Churn by Monthly Charges

Customers paying higher monthly charges are more likely to churn. Very High charges show the highest churn rate at 12%.



Charges Level	Churn Rate
Very Low	~10%
Low	~10%
Medium	~8% (lowest)
High	~11%
Very High	~12% (highest)

9. Key Insights Summary

Based on the exploratory analysis, the following factors are the strongest predictors of customer churn:

#	Insight	Business Action
1	New customers (0-10 months) churn most	Launch onboarding & loyalty programs
2	Month-to-Month contracts = highest churn	Offer discounts to switch to annual plans
3	High complaints = strong churn signal	Resolve complaints within 24 hours
4	Low satisfaction predicts churn	Regular satisfaction surveys + follow-up
5	Very High charges = 12% churn rate	Review pricing for high-charge segments

End of EDA Report